Supplementary Materials: Inference for Cross-Sectional Correlation with Mean Shifts

immediate

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Abstract

This short note is used to fix notations and illustrate the calculations.

Keywords: Change point, Equivariance, Panel Data, Quadratic estimator, Unbiasedness.

1 Main Result

Theorem 1 *Under null hypothesis that ε_i and ξ_i are independent and under some additional mild conditions, we have*

$$\sqrt{n}\hat{\rho} \xrightarrow{D} \mathcal{N}(\rho, V_\rho),$$

where $V_\rho = \frac{3W(\boldsymbol{\theta})}{n\sigma_x^2} + \frac{3W(\boldsymbol{\eta})}{n\sigma_y^2} + 3.5.$

The asymptotic variance V_ρ can be consistently estimated, allowing us to test the correlation coefficient based on Theorem 1. Numerical results demonstrate that our test performs well, whereas the standard Pearson correlation test fails completely when $\boldsymbol{\theta}$ and $\boldsymbol{\eta}$ are not constant. Furthermore, we extend Theorem 1 to the case where $\rho \neq 0$, enabling the construction of a confidence interval.

2 Notations

Consider a pair of time series $(\boldsymbol{X}, \boldsymbol{Y})$ with $\boldsymbol{X} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}$ and $\boldsymbol{Y} = \boldsymbol{\eta} + \boldsymbol{\xi}$, where both mean vectors $\boldsymbol{\theta}$ and $\boldsymbol{\eta}$ are piecewise constant. Let $\boldsymbol{\tau}$ be the vector of locations where at least one of the mean vectors changes its value. Let $L = \min_j\{\tau_{j+1} - \tau_j\}$. To fix ideas, we assume that $\{(\varepsilon_i, \xi_i)\}_{i=1}^n$ are IID and focus on the estimation and inference to the correlation coefficient $\mathrm{corr}(\varepsilon_i, \xi_i)$. Note that a direct approach via sample correlation between $\boldsymbol{X}$ and $\boldsymbol{Y}$ is misleading because of the changes of the mean functions.

Define $\sigma_x^2 = \mathrm{Var}(\varepsilon_i)$, $\sigma_y^2 = \mathrm{Var}(\xi_i)$, $\sigma_{xy} = \rho\sigma_x\sigma_y = \mathrm{Cov}(\varepsilon_i, \xi_i)$. We next sketch the research plan, starting with an unbiased estimator to σ_{xy}.

In a recent paper (Hao et al., 2023), we characterized all equivariant quadratic unbiased estimators for the variance σ_x^2 in the univariate case. As a special case, we showed that $\hat{\sigma}_x^2 = (2T_1 - T_2)/(2n)$ is the unique equivariant quadratic unbiased estimator for σ_x^2 for all $\boldsymbol{\theta} \in \Theta_2$. Define

$$\hat{\sigma}_{xy} = \frac{1}{2n}(2Q_1 - Q_2), \quad \text{where } Q_h = \sum_{i=1}^n \left(X_i - X_{i+h}\right)\left(Y_i - Y_{i+h}\right). \tag{1}$$

Our preliminary result shows that $\hat{\sigma}_{xy}$ is the unique equivariant quadratic unbiased estimator for σ_{xy} for any pair $(\boldsymbol{\theta}, \boldsymbol{\eta})$ with $L = \min_j\{\tau_{j+1} - \tau_j\} \geq 2$.

Naturally, we define $\hat{\rho} = \hat{\sigma}_{xy}/\hat{\sigma}_x\hat{\sigma}_y$.

Recall we define

$$T_h = \sum_{i=1}^n (X_i - X_{i+h})^2, \quad R_h = \sum_{i=1}^n (Y_i - Y_{i+h})^2.$$

We want to show that $(T_1, T_2, R_1, R_2, Q_1, Q_2)^\top/n$ is asymptotically normal. In order to specify the asymptotic distribution, we have to calculate the mean and covariance.

Let $\omega_1^2 = W(\boldsymbol{\theta})/n$ and $\omega_2^2 = W(\boldsymbol{\eta})/n$. By Proposition 2.1 in (Hao et al., 2023), we have $\mathrm{E}(T_h/n) = 2\sigma_x^2 + h\omega_1^2$ and $\mathrm{E}(R_h/n) = 2\sigma_y^2 + h\omega_2^2$. We will complete the rest of the calculation.

We start with $\mathrm{E}(Q_h)$.

$$\mathrm{E}[(X_i - X_{i+h})(Y_i - Y_{i+h})] = (\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) + 2\sigma_{xy} \Rightarrow$$
$$\mathrm{E}(Q_h/n) = 2\sigma_{xy} + h\omega_{12},$$

where $\omega_{12} = W(\boldsymbol{\theta}, \boldsymbol{\eta})/n$, where

$$W(\boldsymbol{\theta}, \boldsymbol{\eta}) = \sum_{i=1}^n (\theta_i - \theta_{i+1})(\eta_i - \eta_{i+1}).$$

Immediately we see $(2Q_1 - Q_2)/2n$ is an unbiased estimator to σ_{xy}.

Next we want to show the asymptotic distribution of

$$\hat{\rho} = \frac{2Q_1 - Q_2}{\sqrt{2T_1 - T_2}\sqrt{2R_1 - R_2}}.$$

We need $\mathrm{Cov}(T_h, R_k)$, $\mathrm{Cov}(T_h, Q_k)$, $\mathrm{Cov}(Q_1, Q_2)$, $\mathrm{Var}(Q_h)$.

Notations. $\mathrm{E}(\varepsilon_i^4) = \kappa_{40}\sigma_x^4$, $\mathrm{E}(\xi_i^4) = \kappa_{04}\sigma_y^4$, $\mathrm{E}(\varepsilon_i^3\xi_i) = \kappa_{31}\sigma_x^3\sigma_y$, $\mathrm{E}(\varepsilon_i\xi_i^3) = \kappa_{13}\sigma_x\sigma_y^3$. $\mathrm{E}(\varepsilon_i^2\xi_i^2) = \kappa_{22}\sigma_x^2\sigma_y^2$.

$\mathrm{E}(\varepsilon_i^3) = \kappa_{30}\sigma_x^3$, $\mathrm{E}(\xi_i^3) = \kappa_{03}\sigma_y^3$, $\mathrm{E}(\varepsilon_i\xi_i^2) = \kappa_{12}\sigma_x\sigma_y^2$, $\mathrm{E}(\varepsilon_i^2\xi_i) = \kappa_{21}\sigma_x^2\sigma_y$.

For a vector $\boldsymbol{v} = (v_1, ..., v_n)^\top$ we can define a family of pseudo norms called tv-norms indexed by positive integers $\|\boldsymbol{v}\|_{tv,h} = \left(\sum_{i=1}^n (v_i - v_{i+h})^2\right)^{\frac{1}{2}}$. It is straightforward to show that $\|\cdot\|_{tv,h}$ satisfies all axioms but the zero condition because $\|\boldsymbol{v}\|_{tv,h} = 0$ does not imply $\boldsymbol{v} = \boldsymbol{0}$. Similarly we can define a pseudo inner product $(\boldsymbol{v}, \boldsymbol{w})_{tv,h} = \sum_{i=1}^n (v_i - v_{i+h})(w_i - w_{i+h})$.

3 Calculation

We will assume $\sigma_x^2 = \sigma_y^2 = 1$ to simplify the notations. If not, we can start with $(\boldsymbol{X}/\sigma_x, \boldsymbol{Y}/\sigma_y)$ which satisfies the assumption.

Let us illustrate how to calculate $\mathrm{Var}(Q_h)$.

$$\begin{aligned}
\mathrm{Var}(Q_h) &= \mathrm{Var}\left(\sum_{i=1}^n (X_i - X_{i+h})(Y_i - Y_{i+h})\right) \\
&= \mathrm{Cov}\left(\sum_{i=1}^n (X_i - X_{i+h})(Y_i - Y_{i+h}), \sum_{i'=1}^n (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right) \\
&= \sum_{i=1}^n \sum_{i'=1}^n \mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right).
\end{aligned}$$

Therefore, we need to calculate

$$\mathrm{Var}\left((X_i - X_{i+h})(Y_i - Y_{i+h})\right),$$

and for $i \neq i'$,

$$\operatorname{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right).$$

To calculate the variance,

$$\begin{aligned}
&\operatorname{Var}\left[(X_i - X_{i+h})(Y_i - Y_{i+h})\right] \\
=&\operatorname{Var}\left[(\theta_i - \theta_{i+h} + \varepsilon_i - \varepsilon_{i+h})(\eta_i - \eta_{i+h} + \xi_i - \xi_{i+h})\right] \\
=&\operatorname{Var}\left[(\theta_i - \theta_{i+h})(\xi_i - \xi_{i+h}) + (\eta_i - \eta_{i+h})(\varepsilon_i - \varepsilon_{i+h}) + (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})\right] \\
=&(\theta_i - \theta_{i+h})^2 \operatorname{Var}(\xi_i - \xi_{i+h}) + (\eta_i - \eta_{i+h})^2 \operatorname{Var}(\varepsilon_i - \varepsilon_{i+h}) + \operatorname{Var}[(\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})] \\
&+ 2(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h})\operatorname{Cov}(\varepsilon_i - \varepsilon_{i+h}, \xi_i - \xi_{i+h}) + 2(\theta_i - \theta_{i+h})\operatorname{Cov}(\xi_i - \xi_{i+h}, (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})) \\
&+ 2(\eta_i - \eta_{i+h})\operatorname{Cov}(\varepsilon_i - \varepsilon_{i+h}, (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})).
\end{aligned}$$

It is easy to show $\operatorname{Var}(\varepsilon_i - \varepsilon_{i+h}) = \operatorname{Var}(\xi_i - \xi_{i+h}) = 2$, $\operatorname{Cov}(\varepsilon_i - \varepsilon_{i+h}, (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})) = \mathrm{E}((\varepsilon_i - \varepsilon_{i+h})^2(\xi_i - \xi_{i+h})) = 0$, $\operatorname{Cov}(\xi_i - \xi_{i+h}, (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})) = 0$, $\operatorname{Cov}(\xi_i - \xi_{i+h}, \varepsilon_i - \varepsilon_{i+h}) = 2\rho$. Now we calculate $\operatorname{Var}[(\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})]$.

$$\begin{aligned}
&\operatorname{Var}[(\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})] \\
=&\mathrm{E}[(\varepsilon_i - \varepsilon_{i+h})^2(\xi_i - \xi_{i+h})^2] - (\mathrm{E}[(\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})])^2 \\
=&\mathrm{E}[(\varepsilon_i^2 + \varepsilon_{i+h}^2 - 2\varepsilon_i\varepsilon_{i+h})(\xi_i^2 + \xi_{i+h}^2 - 2\xi_i\xi_{i+h})] - (2\rho)^2 \\
=&\mathrm{E}[\varepsilon_i^2\xi_i^2 + \varepsilon_i^2\xi_{i+h}^2 + \varepsilon_{i+h}^2\xi_i^2 + \varepsilon_{i+h}^2\xi_{i+h}^2 + 4\varepsilon_i\varepsilon_{i+h}\xi_i\xi_{i+h}] - 4\rho^2 \\
=&2\kappa_{22} + 2.
\end{aligned}$$

Altogether, we have

$$\begin{aligned}
&\operatorname{Var}\left[(X_i - X_{i+h})(Y_i - Y_{i+h})\right] \\
=&2(\theta_i - \theta_{i+h})^2 + 2(\eta_i - \eta_{i+h})^2 + 2\kappa_{22} + 2 + 2(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h})2\rho \\
=&2(\theta_i - \theta_{i+h})^2 + 2(\eta_i - \eta_{i+h})^2 + 2\kappa_{22} + 2 + 4\rho(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}).
\end{aligned}$$

Now let us try

$$\operatorname{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right).$$

$$
\begin{aligned}
&\operatorname{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right) \\
=&\operatorname{Cov}((\theta_i - \theta_{i+h} + \varepsilon_i - \varepsilon_{i+h})(\eta_i - \eta_{i+h} + \xi_i - \xi_{i+h}), (\theta_{i'} - \theta_{i'+h} + \varepsilon_{i'} - \varepsilon_{i'+h})(\eta_{i'} - \eta_{i'+h} + \xi_{i'} - \xi_{i'+h})) \\
=&\operatorname{Cov}((\theta_i - \theta_{i+h})(\xi_i - \xi_{i+h}) + (\eta_i - \eta_{i+h})(\varepsilon_i - \varepsilon_{i+h}) + (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h}), \\
&\qquad (\theta_{i'} - \theta_{i'+h})(\xi_{i'} - \xi_{i'+h}) + (\eta_{i'} - \eta_{i'+h})(\varepsilon_{i'} - \varepsilon_{i'+h}) + (\varepsilon_{i'} - \varepsilon_{i'+h})(\xi_{i'} - \xi_{i'+h})),
\end{aligned}
$$

which is nonzero only when $|i-i'| = h$. As we will need a slightly more general formula to calculate $\operatorname{Cov}(Q_1, Q_2)$, we now find, for three different indices i, i', i'',

$$
\begin{aligned}
&\operatorname{Cov}\left((X_i - X_{i'})(Y_i - Y_{i'}), (X_{i'} - X_{i''})(Y_{i'} - Y_{i''})\right) \\
=&\operatorname{Cov}((\theta_i - \theta_{i'} + \varepsilon_i - \varepsilon_{i'})(\eta_i - \eta_{i'} + \xi_i - \xi_{i'}), (\theta_{i'} - \theta_{i''} + \varepsilon_{i'} - \varepsilon_{i''})(\eta_{i'} - \eta_{i''} + \xi_{i'} - \xi_{i''})) \\
=&\operatorname{Cov}((\theta_i - \theta_{i'})(\xi_i - \xi_{i'}) + (\eta_i - \eta_{i'})(\varepsilon_i - \varepsilon_{i'}) + (\varepsilon_i - \varepsilon_{i'})(\xi_i - \xi_{i'}), \\
&\qquad (\theta_{i'} - \theta_{i''})(\xi_{i'} - \xi_{i''}) + (\eta_{i'} - \eta_{i''})(\varepsilon_{i'} - \varepsilon_{i''}) + (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})) \\
=&(\theta_i - \theta_{i'})(\theta_{i'} - \theta_{i''})\operatorname{Cov}((\xi_i - \xi_{i'}), (\xi_{i'} - \xi_{i''})) + (\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''})\operatorname{Cov}((\xi_i - \xi_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})) \\
&+ (\theta_i - \theta_{i'})\operatorname{Cov}((\xi_i - \xi_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})) \\
&+ (\eta_i - \eta_{i'})(\theta_{i'} - \theta_{i''})\operatorname{Cov}((\varepsilon_i - \varepsilon_{i'}), (\xi_{i'} - \xi_{i''})) + (\eta_i - \eta_{i'})(\eta_{i'} - \eta_{i''})\operatorname{Cov}((\varepsilon_i - \varepsilon_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})) \\
&+ (\eta_i - \eta_{i'})\operatorname{Cov}((\varepsilon_i - \varepsilon_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})) \\
&+ (\theta_{i'} - \theta_{i''})\operatorname{Cov}((\varepsilon_i - \varepsilon_{i'})(\xi_i - \xi_{i'}), (\xi_{i'} - \xi_{i''})) + (\eta_{i'} - \eta_{i''})\operatorname{Cov}((\varepsilon_i - \varepsilon_{i'})(\xi_i - \xi_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})) \\
&+ \operatorname{Cov}((\varepsilon_i - \varepsilon_{i'})(\xi_i - \xi_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})) \\
=&(\theta_i - \theta_{i'})(\theta_{i'} - \theta_{i''})(-1) + (\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''})(-\rho) + (\theta_i - \theta_{i'})(-\kappa_{12}) \\
&+ (\eta_i - \eta_{i'})(\theta_{i'} - \theta_{i''})(-\rho) + (\eta_i - \eta_{i'})(\eta_{i'} - \eta_{i''})(-1) + (\eta_i - \eta_{i'})(-\kappa_{21}) \\
&+ (\theta_{i'} - \theta_{i''})\kappa_{12} + (\eta_{i'} - \eta_{i''})\kappa_{21} + (\kappa_{22} - \rho^2) \\
=&(\kappa_{22} - \rho^2) - (\theta_i - \theta_{i'})(\theta_{i'} - \theta_{i''}) - (\eta_i - \eta_{i'})(\eta_{i'} - \eta_{i''}) \\
&- (\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''})\rho - (\eta_i - \eta_{i'})(\theta_{i'} - \theta_{i''})\rho \\
&- (\theta_i - 2\theta_{i'} + \theta_{i''})\kappa_{12} - (\eta_i - 2\eta_{i'} + \eta_{i''})\kappa_{21}.
\end{aligned}
$$

Now we are ready to calculate

$$
\begin{aligned}
&\mathrm{Var}(Q_h) \\
=&\sum_{i=1}^{n}\sum_{i'=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right) \\
=&\sum_{i=1}^{n}\mathrm{Var}\left((X_i - X_{i+h})(Y_i - Y_{i+h})\right) \\
&+ \sum_{1\le i,i'\le n, i\neq i''}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right) \\
=&\sum_{i=1}^{n}[2(\theta_i - \theta_{i+h})^2 + 2(\eta_i - \eta_{i+h})^2 + 2\kappa_{22} + 2 + 4(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h})\rho] \\
&+ 2\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i+h} - X_{i+2h})(Y_{i+h} - Y_{i+2h})\right) \\
=&2n(\kappa_{22}+1) + 2\sum_{i=1}^{n}(\theta_i - \theta_{i+h})^2 + 2\sum_{i=1}^{n}(\eta_i - \eta_{i+h})^2 + 4\rho\sum_{i=1}^{n}(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) \\
&+ 2\sum_{i=1}^{n}[(\kappa_{22} - \rho^2) - (\theta_i - \theta_{i+h})(\theta_{i+h} - \theta_{i+2h}) - (\eta_i - \eta_{i+h})(\eta_{i+h} - \eta_{i+2h}) \\
&- (\theta_i - \theta_{i+h})(\eta_{i+h} - \eta_{i+2h})\rho - (\eta_i - \eta_{i+h})(\theta_{i+h} - \theta_{i+2h})\rho \\
&- (\theta_i - 2\theta_{i+h} + \theta_{i+2h})\kappa_{12} - (\eta_i - 2\eta_{i+h} + \eta_{i+2h})\kappa_{21}] \\
=&2n(\kappa_{22}+1) + 2hW(\boldsymbol{\theta}) + 2hW(\boldsymbol{\eta}) + 4h\rho W(\boldsymbol{\theta},\boldsymbol{\eta}) + 0 + 0 \\
&+ 2n(\kappa_{22} - \rho^2) - 2\sum_{i=1}^{n}(\theta_i - \theta_{i+h})(\theta_{i+h} - \theta_{i+2h}) - 2\sum_{i=1}^{n}(\eta_i - \eta_{i+h})(\eta_{i+h} - \eta_{i+2h}) \\
&- 2\rho\sum_{i=1}^{n}(\theta_i - \theta_{i+h})(\eta_{i+h} - \eta_{i+2h}) - 2\rho\sum_{i=1}^{n}(\eta_i - \eta_{i+h})(\theta_{i+h} - \theta_{i+2h}) \\
&- 2\kappa_{12}\sum_{i=1}^{n}(\theta_i - 2\theta_{i+h} + \theta_{i+2h}) - 2\kappa_{21}\sum_{i=1}^{n}(\eta_i - 2\eta_{i+h} + \eta_{i+2h}) \\
=&2n(\kappa_{22}+1) + 2hW(\boldsymbol{\theta}) + 2hW(\boldsymbol{\eta}) + 4h\rho W(\boldsymbol{\theta},\boldsymbol{\eta}) + 2n(\kappa_{22} - \rho^2). \\
=&2n(2\kappa_{22} + 1 - \rho^2) + 2hW(\boldsymbol{\theta}) + 2hW(\boldsymbol{\eta}) + 4h\rho W(\boldsymbol{\theta},\boldsymbol{\eta}).
\end{aligned}
$$

The second last equal sign follows $h \le L/2$.

$\mathrm{Cov}(Q_h.Q_k)$ can be calculated similarly.

$$
\begin{aligned}
&\mathrm{Cov}(Q_h.Q_k) \\
=&\sum_{i=1}^{n}\sum_{i'=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i'} - X_{i'+k})(Y_{i'} - Y_{i'+k})\right),
\end{aligned}
$$

where the summands are nonzero only when $i = i'$, or $i = i' + k$, or $i + h = i'$, or $i + h = i' + k$.

$$
\begin{aligned}
&\mathrm{Cov}(Q_h.Q_k) \\
=&\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_i - X_{i+k})(Y_i - Y_{i+k})\right) \\
&+\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i-k} - X_i)(Y_{i-k} - Y_i)\right) \\
&+\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i+h-k} - X_{i+h})(Y_{i+h-k} - Y_{i+h})\right) \\
&+\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i+h} - X_{i+h+k})(Y_{i+h} - Y_{i+h+k})\right) \\
=&4n(\kappa_{22} - \rho^2) + 2hW(\theta) + 2hW(\eta) + 4\rho hW(\theta, \eta).
\end{aligned}
$$

The calculation so far is probably good enough to show Theorem 1. However, if we want to show a confidence interval, we have to derive the full covariance matrix.

4 Additional calculation for general ρ

Warning: the following calculation should be essentially correct but may contain errors or typos.

Now consider $\mathrm{Cov}(T_h, R_h)$.

$$
\begin{aligned}
&\mathrm{Cov}(T_h, R_h)\\
=&\mathrm{Cov}\left(\sum_{i=1}^{n}(X_i - X_{i+h})^2, \sum_{i'=1}^{n}(Y_{i'} - Y_{i'+h})^2\right)\\
=&\sum_{i=1}^{n}\sum_{i'=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i'} - Y_{i'+h})^2\right)\\
=&\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_i - Y_{i+h})^2\right) + \sum_{1\le i,i'\le n, i\neq i'}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i'} - Y_{i'+h})^2\right)\\
=&\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_i - Y_{i+h})^2\right) + 2\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i+h} - Y_{i+2h})^2\right)
\end{aligned}
$$

Let us calculate $\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_i - Y_{i+h})^2\right)$ and $\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i+h} - Y_{i+2h})^2\right)$ respectively.

$$
\begin{aligned}
&\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_i - Y_{i+h})^2\right)\\
=&\mathrm{Cov}\left((\theta_i - \theta_{i+h} + \varepsilon_i - \varepsilon_{i+h})^2, (\eta_i - \eta_{i+h} + \xi_i - \xi_{i+h})^2\right)\\
=&\mathrm{Cov}\left(2(\theta_i - \theta_{i+h})(\varepsilon_i - \varepsilon_{i+h}) + (\varepsilon_i - \varepsilon_{i+h})^2, 2(\eta_i - \eta_{i+h})(\xi_i - \xi_{i+h}) + (\xi_i - \xi_{i+h})^2\right)\\
=&4(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h})\mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i+h}), (\xi_i - \xi_{i+h})\right) + \mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i+h})^2, (\xi_i - \xi_{i+h})^2\right)\\
&+ 2(\theta_i - \theta_{i+h})\mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i+h}), (\xi_i - \xi_{i+h})^2\right) + 2(\eta_i - \eta_{i+h})\mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i+h})^2, (\xi_i - \xi_{i+h})\right)\\
=&4(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h})(2\rho) + (2(\kappa_{22} - 1) + 4\rho^2) + 2(\theta_i - \theta_{i+h})0 + 2(\eta_i - \eta_{i+h})0\\
=&8\rho(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) + 2(\kappa_{22} - 1) + 4\rho^2
\end{aligned}
$$

For $\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i+h} - Y_{i+2h})^2\right)$, we calculate a slightly more general one: for three different indices i, i' and i''

$$
\begin{aligned}
& \mathrm{Cov}\left((X_i - X_{i'})^2, (Y_{i'} - Y_{i''})^2\right) \\
=& \mathrm{Cov}((\theta_i - \theta_{i'} + \varepsilon_i - \varepsilon_{i'})^2, (\eta_{i'} - \eta_{i''} + \xi_{i'} - \xi_{i''})^2) \\
=& \mathrm{Cov}(2(\theta_i - \theta_{i'})(\varepsilon_i - \varepsilon_{i'}) + (\varepsilon_i - \varepsilon_{i'})^2, 2(\eta_{i'} - \eta_{i''})(\xi_{i'} - \xi_{i''}) + (\xi_{i'} - \xi_{i''})^2) \\
=& 4(\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''})\mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i'}), (\xi_{i'} - \xi_{i''})\right) + 2(\theta_i - \theta_{i'})\mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i'}), (\xi_{i'} - \xi_{i''})^2\right) \\
& + 2(\eta_{i'} - \eta_{i''})\mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i'})^2, (\xi_{i'} - \xi_{i''})\right) + \mathrm{Cov}\left((\varepsilon_i - \varepsilon_{i'})^2, (\xi_{i'} - \xi_{i''})^2\right) \\
=& 4(\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''})(-\rho) + 2(\theta_i - \theta_{i'})(-\kappa_{12}) + 2(\eta_{i'} - \eta_{i''})(\kappa_{21}) + (\kappa_{22} - 1) \\
=& \kappa_{22} - 1 - 4\rho(\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''}) - 2\kappa_{12}(\theta_i - \theta_{i'}) + 2\kappa_{21}(\eta_{i'} - \eta_{i''})
\end{aligned}
$$

Together,

$$
\begin{aligned}
& \mathrm{Cov}(T_h, R_h) \\
=& \sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_i - Y_{i+h})^2\right) + 2\sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i+h} - Y_{i+2h})^2\right) \\
=& \sum_{i=1}^{n} \left(8\rho(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) + 2(\kappa_{22} - 1) + 4\rho^2\right) \\
& + 2\sum_{i=1}^{n} (\kappa_{22} - 1 - 4\rho(\theta_i - \theta_{i+h})(\eta_{i+h} - \eta_{i+2h}) - 2\kappa_{12}(\theta_i - \theta_{i+h}) - 2\kappa_{21}(\eta_{i+h} - \eta_{i+2h})) \\
=& 2n(\kappa_{22} - 1) + 4n\rho^2 + 8\rho\sum_{i=1}^{n}(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) \\
& + 2n(\kappa_{22} - 1) - 8\rho\sum_{i=1}^{n}(\theta_i - \theta_{i+h})(\eta_{i+h} - \eta_{i+2h}) - 4\kappa_{12}\sum_{i=1}^{n}(\theta_i - \theta_{i+h}) - 4\kappa_{21}\sum_{i=1}^{n}(\eta_{i+h} - \eta_{i+2h}) \\
=& 4n(\kappa_{22} - 1) + 4n\rho^2 + 8\rho h W(\boldsymbol{\theta}, \boldsymbol{\eta}).
\end{aligned}
$$

Similarly, you can find $\mathrm{Cov}(T_h, R_k)$, $(h \neq k)$.

$$
\begin{aligned}
&\mathrm{Cov}(T_h, R_k)\\
=&\mathrm{Cov}\left(\sum_{i=1}^{n}(X_i - X_{i+h})^2, \sum_{i'=1}^{n}(Y_{i'} - Y_{i'+k})^2\right)\\
=&\sum_{i=1}^{n}\sum_{i'=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i'} - Y_{i'+k})^2\right)\\
=&\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_i - Y_{i+k})^2\right) + \sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i+h} - Y_{i+h+k})^2\right)\\
&+\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i-k} - Y_i)^2\right) + \sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (Y_{i+h-k} - Y_{i+h})^2\right)\\
=&4n(\kappa_{22} - 1) + 8\rho W(\theta, \eta). \quad \text{double check this}
\end{aligned}
$$

Now consider $\mathrm{Cov}(T_h, Q_h)$.

$$
\begin{aligned}
&\mathrm{Cov}(T_h, Q_h)\\
=&\mathrm{Cov}\left(\sum_{i=1}^{n}(X_i - X_{i+h})^2, \sum_{i'=1}^{n}(X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right)\\
=&\sum_{i=1}^{n}\sum_{i'=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right)\\
=&\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_i - X_{i+h})(Y_i - Y_{i+h})\right) + \sum_{1\le i,i'\le n, i\ne i'}\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_{i'} - X_{i'+h})(Y_{i'} - Y_{i'+h})\right)\\
=&\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_i - X_{i+h})(Y_i - Y_{i+h})\right) + 2\sum_{i=1}^{n}\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_{i+h} - X_{i+2h})(Y_{i+h} - Y_{i+2h})\right).
\end{aligned}
$$

Now we calculate $\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_i - X_{i+h})(Y_i - Y_{i+h})\right)$ and $\mathrm{Cov}\left((X_i - X_{i+h})^2, (X_{i+h} - X_{i+2h})(Y_{i+h} - Y_{i+2h})\right)$ respectively.

$$
\begin{aligned}
&\operatorname{Cov}\left((X_i - X_{i+h})^2, (X_i - X_{i+h})(Y_i - Y_{i+h})\right) \\
=&\operatorname{Cov}\left((\theta_i - \theta_{i+h} + \varepsilon_i - \varepsilon_{i+h})^2, (\theta_i - \theta_{i+h} + \varepsilon_i - \varepsilon_{i+h})(\eta_i - \eta_{i+h} + \xi_i - \xi_{i+h})\right) \\
=&\operatorname{Cov}\left(2(\theta_i - \theta_{i+h})(\varepsilon_i - \varepsilon_{i+h}) + (\varepsilon_i - \varepsilon_{i+h})^2, (\theta_i - \theta_{i+h})(\xi_i - \xi_{i+h}) + (\eta_i - \eta_{i+h})(\varepsilon_i - \varepsilon_{i+h}) + (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})\right. \\
=&2(\theta_i - \theta_{i+h})^2 \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i+h}), (\xi_i - \xi_{i+h})\right) + 2(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) \operatorname{Var}(\varepsilon_i - \varepsilon_{i+h}) \\
&+ 2(\theta_i - \theta_{i+h}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i+h}), (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})\right) + (\theta_i - \theta_{i+h}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i+h})^2, (\xi_i - \xi_{i+h})\right) \\
&+ (\eta_i - \eta_{i+h}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i+h})^2, (\varepsilon_i - \varepsilon_{i+h})\right) + \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i+h})^2, (\varepsilon_i - \varepsilon_{i+h})(\xi_i - \xi_{i+h})\right) \\
=&2(\theta_i - \theta_{i+h})^2(2\rho) + 2(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h})(2) + 2(\theta_i - \theta_{i+h})(0) + (\theta_i - \theta_{i+h})(0) + (\eta_i - \eta_{i+h})(0) + 2 + 2\kappa_{31} \\
=&4\rho(\theta_i - \theta_{i+h})^2 + 4(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) + 2\rho + 2\kappa_{31}.
\end{aligned}
$$

For three different indices i, i' and i''

$$
\begin{aligned}
&\operatorname{Cov}\left((X_i - X_{i'})^2, (X_{i'} - X_{i''})(Y_{i'} - Y_{i''})\right) \\
=&\operatorname{Cov}\left((\theta_i - \theta_{i'} + \varepsilon_i - \varepsilon_{i'})^2, (\theta_{i'} - \theta_{i''} + \varepsilon_{i'} - \varepsilon_{i''})(\eta_{i'} - \eta_{i''} + \xi_{i'} - \xi_{i''})\right) \\
=&\operatorname{Cov}\left(2(\theta_i - \theta_{i'})(\varepsilon_i - \varepsilon_{i'}) + (\varepsilon_i - \varepsilon_{i'})^2, (\theta_{i'} - \theta_{i''})(\xi_{i'} - \xi_{i''}) + (\eta_{i'} - \eta_{i''})(\varepsilon_{i'} - \varepsilon_{i''}) + (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})\right) \\
=&2(\theta_i - \theta_{i'})(\theta_{i'} - \theta_{i''}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i'}), (\xi_{i'} - \xi_{i''})\right) + 2(\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''}) \operatorname{Cov}\left(\varepsilon_i - \varepsilon_{i'}, \varepsilon_{i'} - \varepsilon_{i''}\right) \\
&+ 2(\theta_i - \theta_{i'}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i'}), (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})\right) + (\theta_{i'} - \theta_{i''}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i'})^2, (\xi_{i'} - \xi_{i''})\right) \\
&+ (\eta_{i'} - \eta_{i''}) \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i'})^2, (\varepsilon_{i'} - \varepsilon_{i''})\right) + \operatorname{Cov}\left((\varepsilon_i - \varepsilon_{i'})^2, (\varepsilon_{i'} - \varepsilon_{i''})(\xi_{i'} - \xi_{i''})\right) \\
=&2(\theta_i - \theta_{i'})(\theta_{i'} - \theta_{i''})(-\rho) + 2(\theta_i - \theta_{i'})(\eta_{i'} - \eta_{i''})(-1) + 2(\theta_i - \theta_{i'})(-\kappa_{21}) \\
&+ (\theta_{i'} - \theta_{i''})\kappa_{21} + (\eta_{i'} - \eta_{i''})\kappa_{30} + \kappa_{31} - \rho.
\end{aligned}
$$

$$
\begin{aligned}
&\operatorname{Cov}(T_h, Q_h) \\
=&\sum_{i=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_i - X_{i+h})(Y_i - Y_{i+h})\right) + 2\sum_{i=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_{i+h} - X_{i+2h})(Y_{i+h} - Y_{i+2h})\right). \\
=&\sum_{i=1}^{n} \left(4\rho(\theta_i - \theta_{i+h})^2 + 4(\theta_i - \theta_{i+h})(\eta_i - \eta_{i+h}) + 2\rho + 2\kappa_{31}\right) \\
&+ 2\sum_{i=1}^{n}[2(\theta_i - \theta_{i+h})(\theta_{i+h} - \theta_{i+2h})(-\rho) + 2(\theta_i - \theta_{i+h})(\eta_{i+h} - \eta_{i+2h})(-1) + 2(\theta_i - \theta_{i+h})(-\kappa_{21}) \\
&+ (\theta_{i+h} - \theta_{i+2h})\kappa_{21} + (\eta_{i+h} - \eta_{i+2h})\kappa_{3,X} + \kappa_{31} - \rho] \\
=&4\rho h W(\boldsymbol{\theta}) + 4hW(\boldsymbol{\theta}, \boldsymbol{\eta}) + 2n(3\rho + \kappa_{31}) + 2n(\kappa_{31} - \rho) \\
=&4\rho h W(\boldsymbol{\theta}) + 4hW(\boldsymbol{\theta}, \boldsymbol{\eta}) + 4n\kappa_{31}
\end{aligned}
$$

Similarly we can calculate $\operatorname{Cov}(T_h, Q_k)$.

$$
\begin{aligned}
&\operatorname{Cov}(T_h, Q_k) \\
=&\operatorname{Cov}\left(\sum_{i=1}^{n}(X_i - X_{i+h})^2, \sum_{i'=1}^{n}(X_{i'} - X_{i'+k})(Y_{i'} - Y_{i'+k})\right) \\
=&\sum_{i=1}^{n}\sum_{i'=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_{i'} - X_{i'+k})(Y_{i'} - Y_{i'+k})\right) \\
=&\sum_{i=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_i - X_{i+k})(Y_i - Y_{i+k})\right) + \sum_{i=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_{i+h} - X_{i+h+k})(Y_{i+h} - Y_{i+h+k})\right) \\
&+ \sum_{i=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_{i-k} - X_i)(Y_{i-k} - Y_i)\right) + \sum_{i=1}^{n} \operatorname{Cov}\left((X_i - X_{i+h})^2, (X_{i-k+h} - X_{i+h})(Y_{i-k+h} - Y_{i+h})\right) \\
=&4n(\kappa_{31} - \rho) + 4\rho W(\boldsymbol{\theta}) + 4W(\boldsymbol{\theta}, \boldsymbol{\eta}).
\end{aligned}
$$

5 Asymptotic normality

We will modify the following proof.

Lemma 1 *Under conditions 1, 2, and 4, and the null hypothesis that $\varepsilon_1, \ldots, \varepsilon_n$ are IID, if $K \leq L/2$, then $\boldsymbol{T}_K = (T_1, ..., T_K)^\top$ is asymptotically normal as $n \to \infty$,*

$$\boldsymbol{\Sigma}_{K,w}^{-1/2} \sqrt{n}(\boldsymbol{T}_K/n - \boldsymbol{\nu}_K) \xrightarrow{D} \mathcal{N}(\mathbf{0}_K, \boldsymbol{I}_K), \tag{2}$$

where

$$\boldsymbol{\Sigma}_{K,w} = 4\gamma_0^2 \left[\boldsymbol{I}_K + (\kappa_4 - 1)\mathbf{1}_K \mathbf{1}_K^\top + 2w\boldsymbol{H}_K\right], \quad \boldsymbol{\nu}_K = \gamma_0(2 \cdot \mathbf{1}_K + w\boldsymbol{\eta}_K).$$

Moreover, the range of K can be relaxed to $K \leq L$, in which case

$$\lim_{n\to\infty} \boldsymbol{\Sigma}_{K,2w}^{-1/2} \sqrt{n}(\boldsymbol{T}_K/n - \boldsymbol{\nu}_K) \preceq \mathcal{N}(\mathbf{0}_K, \boldsymbol{I}_K). \tag{3}$$

Proof of Lemma 1. It follows Proposition 2.1 in Hao et al. (2023) directly that the expectation and covariance of $n^{-1}\boldsymbol{T}_K$ are $\boldsymbol{\nu}_K$ and $n^{-1}\boldsymbol{\Sigma}_K$, respectively. We only need to show that $\boldsymbol{T}_K$ is jointly normal asymptotically. For a marginal T_k with $1 \leq k \leq K$,

$$\begin{aligned}
T_k &= \sum_{i=1}^n (X_i - X_{i+k})^2 \\
&= \sum_{i=1}^n (\theta_i - \theta_{i+k} + \varepsilon_i - \varepsilon_{i+k})^2 \\
&= \sum_{i=1}^n (\theta_i - \theta_{i+k})^2 + 2\sum_{i=1}^n (\theta_i - \theta_{i+k})(\varepsilon_i - \varepsilon_{i+k}) + \sum_{i=1}^n (\varepsilon_i - \varepsilon_{i+k})^2 \\
&= 2n\sigma^2 + kW(\boldsymbol{\theta}) + 2\sum_{i=1}^n (\theta_i - \theta_{i+k})(\varepsilon_i - \varepsilon_{i+k}) + \sum_{i=1}^n \left[(\varepsilon_i - \varepsilon_{i+k})^2 - 2\sigma^2\right],
\end{aligned}$$

where $\sum_{i=1}^n \left[(\varepsilon_i - \varepsilon_{i+k})^2 - 2\sigma^2\right]$ is asymptotically normal as it is a sum of weakly dependent and identically distributed random variables. For the term $2\sum_{i=1}^n (\theta_i - \theta_{i+k})(\varepsilon_i - \varepsilon_{i+k})$, the set of coefficients $\{\theta_i - \theta_{i+k}\}$ consists of k copies of the values $\mu_1 - \mu_2$,..., $\mu_{J+1} - \mu_1$ and $n - (J+1)k$ zeros. So the Euclidean norm of the coefficient vector is $\sqrt{kW(\boldsymbol{\theta})}$. When $W(\boldsymbol{\theta}) = o(n)$, this term is dominated by the last term and won't express in the limiting distribution. Moreover, if the coefficient vector satisfies Lindeberg condition, i.e. $\max_j\{(\mu_j - \mu_{j+1})^2\}/W(\boldsymbol{\theta}) \to 0$, then $2\sum_{i=1}^n (\theta_i - \theta_{i+k})(\varepsilon_i - \varepsilon_{i+k}) = 2\sum_{i=1}^n (\theta_i - \theta_{i+k})\varepsilon_i - 2\sum_{i=1}^n (\theta_i - \theta_{i+k})\varepsilon_{i+k}$ is a sum of two terms both of which are asymptotically normal. Note that our condition $\max_j\{(\mu_j - \mu_{j+1})^2\} = o(n)$ implies that either $W(\boldsymbol{\theta}) = o(n)$ or $\max_j\{(\mu_j - \mu_{j+1})^2\}/W(\boldsymbol{\theta}) \to 0$. Similarly, we can show that any linear combination of T_k's are asymptotically normal. By the Cramér-Wold device, $\boldsymbol{T}_K$ is asymptotically normal. $\square$

6 Delta method

Given the asymptotic normality of the joint vector $(T_1, T_2, R_1, R_2, Q_1, Q_2)^\top$, we derive the asymptotic property of

$$\hat{\rho} = \frac{2Q_1 - Q_2}{\sqrt{2T_1 - T_2}\sqrt{2R_1 - R_2}}.$$

To facilitate our calculation, we use the notation $\boldsymbol{Z} = (Z_1, \cdots, Z_6)^\top = (T_1, T_2, R_1, R_2, Q_1, Q_2)^\top/(2n)$.

We have $\sqrt{n}(\boldsymbol{Z} - \boldsymbol{\mu}_Z) \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_Z)$, where

$$\boldsymbol{\mu}_Z = \begin{pmatrix} \sigma_1^2 + \frac{1}{2}\omega_1^2 \\ \sigma_1^2 + \omega_1^2 \\ \sigma_2^2 + \frac{1}{2}\omega_2^2 \\ \sigma_2^2 + \omega_2^2 \\ \sigma_{12} + \frac{1}{2}\omega_{12} \\ \sigma_{12} + \omega_{12} \end{pmatrix} = \begin{pmatrix} 1 + \frac{1}{2}\omega_1^2 \\ 1 + \omega_1^2 \\ 1 + \frac{1}{2}\omega_2^2 \\ 1 + \omega_2^2 \\ \rho + \frac{1}{2}\omega_{12} \\ \rho + \omega_{12} \end{pmatrix}.$$

$\hat{\rho} = g(\boldsymbol{Z}) = \frac{2Z_5 - Z_6}{\sqrt{2Z_1 - Z_2}\sqrt{2Z_3 - Z_4}}$.

$$\boldsymbol{\Sigma}_Z = \frac{1}{n}\begin{pmatrix} \kappa_{40} + 2\omega_1^2 & \kappa_{40} - 1 + 2\omega_1^2 & \kappa_{22} - 1 + \rho^2 + 2\rho\omega_{12} & \kappa_{22} - 1 + 2\rho\omega_{12} & \kappa_{31} + \rho\omega_1^2 + \omega_{12} & \kappa_{31} - \rho + \rho\omega_1^2 + \omega_{12} \\ & \kappa_{40} + 4\omega_1^2 & \kappa_{22} - 1 + 2\rho\omega_{12} & \kappa_{22} - 1 + \rho^2 + 4\rho\omega_{12} & \kappa_{31} - \rho + \rho\omega_1^2 + \omega_{12} & \kappa_{31} + 2\rho\omega_1^2 + 2\omega_{12} \\ & & \kappa_{04} + 2\omega_2^2 & \kappa_{04} - 1 + 2\omega_2^2 & \kappa_{13} + \rho\omega_2^2 + \omega_{12} & \kappa_{13} - \rho + \rho\omega_2^2 + \omega_{12} \\ & & & \kappa_{04} + 4\omega_2^2 & \kappa_{13} - \rho + \rho\omega_2^2 + \omega_{12} & \kappa_{13} + 2\rho\omega_2^2 + 2\omega_{12} \\ & & & & \kappa_{22} + \frac{1-\rho^2}{2} + \frac{\omega_1^2 + \omega_2^2 + 2\rho\omega_{12}}{2} & \kappa_{22} - \rho^2 + \frac{\omega_1^2 + \omega_2^2 + 2\rho\omega_{12}}{2} \\ & & & & & \kappa_{22} + \frac{1-\rho^2}{2} + \omega_1^2 + \omega_2^2 + 2\rho\omega_{12} \end{pmatrix}.$$

$$g(\boldsymbol{Z}) = \frac{2Z_5 - Z_6}{\sqrt{2Z_1 - Z_2}\sqrt{2Z_3 - Z_4}}.$$

$$\nabla g(\boldsymbol{\mu}_Z) = \begin{pmatrix} -\frac{1}{2Z_1-Z_2} g(\boldsymbol{Z}) \\ \frac{1}{2}\frac{1}{2Z_1-Z_2} g(\boldsymbol{Z}) \\ -\frac{1}{2Z_3-Z_4} g(\boldsymbol{Z}) \\ \frac{1}{2}\frac{1}{2Z_3-Z_4} g(\boldsymbol{Z}) \\ \frac{2}{2Z_5-Z_6} g(\boldsymbol{Z}) \\ -\frac{1}{2Z_5-Z_6} g(\boldsymbol{Z}) \end{pmatrix}_{\boldsymbol{Z}=\boldsymbol{\mu}_Z} = \begin{pmatrix} -\rho \\ \frac{1}{2}\rho \\ -\rho \\ \frac{1}{2}\rho \\ 2 \\ -1 \end{pmatrix}.$$

Finally,

$$\begin{aligned}(\nabla g(\boldsymbol{\mu}_Z))^\top \boldsymbol{\Sigma}_Z \nabla g(\boldsymbol{\mu}_Z) = \frac{1}{n}\Big[&(1-\rho^2)(\omega_1^2+\omega_2^2-2\rho\omega_{12}) + \frac{\rho^2}{4}(\kappa_{40}+2\kappa_{22}+\kappa_{04}) \\ &-\rho(\kappa_{31}+\kappa_{13}) + \frac{5}{2}(1-\rho^2)^2 + \kappa_{22}\Big]\end{aligned}$$

When $\rho = 0$, this reduces to $\frac{1}{n}(\omega_1^2+\omega_2^2+\kappa_{22}+\frac{5}{2})$, which further reduces to $\frac{1}{n}(\omega_1^2+\omega_2^2+\frac{7}{2})$ under Gaussian assumption.

For general ρ, under Gaussian assumption ($\kappa_{40}=\kappa_{04}=3$, $\kappa_{13}=\kappa_{31}=3\rho$, $\kappa_{22}=1+2\rho^2$), this reduces to $\frac{1}{n}(1-\rho^2)(\omega_1^2+\omega_2^2-2\rho\omega_{12}+\frac{7}{2}(1-\rho^2))$.

$$\begin{aligned}&\sum_{i=1}^n (X_i - X_{i+1})^2 (Y_i - Y_{i+1})^2 \\ =&\sum_{i=1}^n [(\theta_i-\theta_{i+1})^2 + (\varepsilon_i-\varepsilon_{i+1})^2 + 2(\theta_i-\theta_{i+1})(\varepsilon_i-\varepsilon_{i+1})][(\eta_i-\eta_{i+1})^2 + (\xi_i-\xi_{i+1})^2 + 2(\eta_i-\eta_{i+1})(\xi_i-\xi_{i+1})] \\ =&\sum_{i=1}^n [(\theta_i-\theta_{i+1})^2(\eta_i-\eta_{i+1})^2 + (\varepsilon_i-\varepsilon_{i+1})^2(\eta_i-\eta_{i+1})^2 + 2(\theta_i-\theta_{i+1})(\varepsilon_i-\varepsilon_{i+1})(\eta_i-\eta_{i+1})^2] \\ &+ \sum_{i=1}^n [(\theta_i-\theta_{i+1})^2(\xi_i-\xi_{i+1})^2 + (\varepsilon_i-\varepsilon_{i+1})^2(\xi_i-\xi_{i+1})^2 + 2(\theta_i-\theta_{i+1})(\varepsilon_i-\varepsilon_{i+1})(\xi_i-\xi_{i+1})^2] \\ &+ \sum_{i=1}^n [2(\theta_i-\theta_{i+1})^2(\eta_i-\eta_{i+1})(\xi_i-\xi_{i+1}) + 2(\varepsilon_i-\varepsilon_{i+1})^2(\eta_i-\eta_{i+1})(\xi_i-\xi_{i+1}) \\ &+ 4(\theta_i-\theta_{i+1})(\varepsilon_i-\varepsilon_{i+1})(\eta_i-\eta_{i+1})(\xi_i-\xi_{i+1})]\end{aligned}$$

$$
\begin{aligned}
&\mathrm{E}\left(\sum_{i=1}^{n}(X_i - X_{i+1})^2(Y_i - Y_{i+1})^2\right) \\
=&\sum_{i=1}^{n}(\theta_i - \theta_{i+1})^2(\eta_i - \eta_{i+1})^2 + \mathrm{E}(\varepsilon_i - \varepsilon_{i+1})^2\sum_{i=1}^{n}(\eta_i - \eta_{i+1})^2 + 0 \\
&+ \mathrm{E}(\xi_i - \xi_{i+1})^2\sum_{i=1}^{n}(\theta_i - \theta_{i+1})^2 + n\mathrm{E}[(\varepsilon_i - \varepsilon_{i+1})^2(\xi_i - \xi_{i+1})^2] + 0 \\
&+ 0 + 0 + 4\mathrm{E}[(\varepsilon_i - \varepsilon_{i+1})(\xi_i - \xi_{i+1})]\sum_{i=1}^{n}(\theta_i - \theta_{i+1})(\eta_i - \eta_{i+1}) \\
=&\sum_{i=1}^{n}(\theta_i - \theta_{i+1})^2(\eta_i - \eta_{i+1})^2 + 2\sigma_1^2 W(\boldsymbol{\eta}) + 2\sigma_2^2 W(\boldsymbol{\theta}) + n\sigma_1^2\sigma_2^2(2\kappa_{22} - 2 + 4\rho^2) + 8\sigma_{12}W(\boldsymbol{\theta}, \boldsymbol{\eta})
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
&\mathrm{E}\left(\sum_{i=1}^{n}(X_i - X_{i+h})^2(Y_i - Y_{i+h})^2\right) \\
=&n\sigma_1^2\sigma_2^2(2\kappa_{22} - 2 + 4\rho^2) + h\left(\sum_{i=1}^{n}(\theta_i - \theta_{i+1})^2(\eta_i - \eta_{i+1})^2 + 2\sigma_1^2 W(\boldsymbol{\eta}) + 2\sigma_2^2 W(\boldsymbol{\theta}) + 8\sigma_{12}W(\boldsymbol{\theta}, \boldsymbol{\eta})\right)
\end{aligned}
$$

References

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